

MAT111 Assignment 2

Submission: [link](#)

Deadline: 17th, January, 11:59 pm

Disprove for 1 to 5.

Problem 1. *Let n^2 is an even number. Can n be odd?*

Problem 2. *A number is divisible by 6 and 8. Prove that the number is divisible by 48.*

Problem 3. *Prove that the sum of two irrational numbers is always irrational.*

Problem 4. *Let f be a continuous function. Prove that if $f(x_1) = f(x_2)$ then $x_1 = x_2$.*

Problem 5. *Let a function $f : \mathbb{R} \rightarrow \mathbb{R}$ be continuous and satisfies the condition*

$$f(x + y) = f(x) + f(y)$$

Prove that for all $x \in \mathbb{R}$, $f(x) = x$.

Use Contrapositive for 6 to 13

Problem 6. *Let $x, y, z \in \mathbb{Z}$ and $x \neq 0$. Prove that if x does not divide yz then x divides neither y nor z .*

Problem 7. *Use contrapositive to prove that if $n \in \mathbb{N}$ and $2^n - 1$ is prime, then n is prime.*

Problem 8. *If $a \equiv b \pmod{n}$, then $\gcd(a, n) = \gcd(b, n)$.*

Problem 9. *Suppose $x, y, z \in \mathbb{Z}$ and $x \neq 0$. If $x \mid yz$, then $x \mid y$ and $x \mid z$.*

Problem 10. *Suppose $x, y \in \mathbb{Z}$. If $x^2(y + 3)$ is even, then x is even or y is odd.*

Problem 11. *Suppose $a \in \mathbb{Z}$. If a^2 is not divisible by 4, then a is odd.*

Problem 12. *Suppose $x \in \mathbb{R}$. If*

$$x^5 + 7x^3 + 5x \geq x^4 + x^2 + 8,$$

then $x \geq 0$.

Problem 13. *Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a constant function. Prove that $f(x) = f(y)$ for all $x, y \in \mathbb{R}$.*

Use Contradiction for 14 to 22

Problem 14. *Prove that there is no largest even number.*

Problem 15. *For any set A, B , we have $A \cap (B - A) = \emptyset$*

Problem 16. *The sum of an even integer and an odd integer cannot be even.*

Problem 17. *A statement cannot be both true and false at the same time.*

Problem 18. *Prove that if n is odd, then n^2 must be odd too.*

Problem 19. *Prove that there exists no positive integer pair of (a, b) such that*

$$a^2 = 2b^2$$

Problem 20. *Prove that $\sqrt[3]{2}$ is irrational.*

Problem 21. *Prove that $\sqrt{6}$ is irrational.*

Problem 22. *We say that a point $P = (x, y) \in \mathbb{R}^2$ is rational if both x and y are rational. More precisely, P is rational if $P = (x, y) \in \mathbb{Q}^2$.*

An equation $F(x, y) = 0$ is said to have a rational point if there exist $x_0, y_0 \in \mathbb{Q}$ such that $F(x_0, y_0) = 0$. For example, the curve

$$x^2 + y^2 - 1 = 0$$

has the rational point $(x_0, y_0) = (1, 0)$.

Show that the curve

$$x^2 + y^2 - 3 = 0$$

has no rational points. Use this fact to show that, $\sqrt{3}$ is irrational.

Proofs in function for 23 to 27

Problem 23. *Show that the function*

$$g : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z} \times \mathbb{Z}$$

defined by

$$g(m, n) = (m + n, m + 2n)$$

is both injective and surjective.

Problem 24. *Consider the function*

$$\theta : \{0, 1\} \times \mathbb{N} \rightarrow \mathbb{Z}$$

defined by

$$\theta(a, b) = a - 2ab + b.$$

Is θ injective? Is it surjective? Is it bijective? Explain.

Problem 25. Consider the function

$$\theta : \{0, 1\} \times \mathbb{N} \rightarrow \mathbb{Z}$$

defined by

$$\theta(a, b) = (-1)^a b.$$

Is θ injective? Is it surjective? Is it bijective? Explain.

Problem 26. Prove that the function

$$f : \mathbb{N} \rightarrow \mathbb{Z}$$

defined by

$$f(n) = \frac{(-1)^n(2n-1)+1}{4}$$

is bijective.

Problem 27. Prove that the function

$$f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$$

defined by

$$f(m, n) = 2^{m-1}(2n-1)$$

is bijective.

Use Induction for 28 to 32

Problem 28. Use induction to prove that if A is a set and $|A| = n$, then

$$|\mathcal{P}(A)| = 2^n.$$

Problem 29. Suppose $x \in \mathbb{R}$ with $x > -1$. Prove that, for every $n \in \mathbb{N}$,

$$1 + nx \leq (1 + x)^n.$$

Problem 30. Prove that, for every $n \in \mathbb{N}$, there are n distinct natural numbers $a_1, a_2, \dots, a_n$ such that

$$a_1^2 + a_2^2 + \dots + a_n^2$$

is a perfect square.

Problem 31. The binary representation of the number 13 is 1101, because

$$13 = 1 \cdot 2^3 + 1 \cdot 2^2 + 0 \cdot 2^1 + 1 \cdot 2^0.$$

In general, the binary representation of a natural number n is a sequence of digits

$$a_k a_{k-1} a_{k-2} \dots a_2 a_1 a_0$$

where each $a_i \in \{0, 1\}$ and

$$n = a_k \cdot 2^k + a_{k-1} \cdot 2^{k-1} + \dots + a_2 \cdot 2^2 + a_1 \cdot 2^1 + a_0 \cdot 2^0.$$

Prove that every $n \in \mathbb{N}$ has a binary representation.

Problem 32. Suppose $A_1, A_2, A_3, \dots, A_n$ are subsets of some universal set U . Prove that the following hold for every $n \in \mathbb{N}$. You may use the fact that unions and intersections are associative.

1. $(A_1 \cap A_2 \cap A_3 \cap \dots \cap A_n)^c = A_1^c \cup A_2^c \cup A_3^c \cup \dots \cup A_n^c.$
2. $(A_1 \cup A_2 \cup A_3 \cup \dots \cup A_n)^c = A_1^c \cap A_2^c \cap A_3^c \cap \dots \cap A_n^c.$